

Final Term Practice sheet:

Write a program to convert following code into assembly program.

```
IF AX < 0
THEN
PUT -1 IN BX
END IF
```

1.

2.

Write a program to convert following code into assembly program.

```
IF "AL < 0
THEN
put FFh in AH'
ELSE
put 0 in AH
END_IF
```

3.

Write a program to convert following code into assembly program.

Suppose DL contains the ASCII code of a character.

```
(IF DL >= "A") AND (DL <= 'Z')
THEN
    display DL
END_IF
```

4.

Write a program to convert following code into assembly program.

```
IF AX < BX
THEN
  IF BX < CX
  THEN
    put 0 in AX
  ELSE
    put 0 in BX
  END_IF
END_IF
```

5.

Write a program to convert following code into assembly program.

```
IF (AX < BX) OR (BX < CX)
THEN
  put 0 in DX
ELSE
  put 1 in DX
END_IF
```

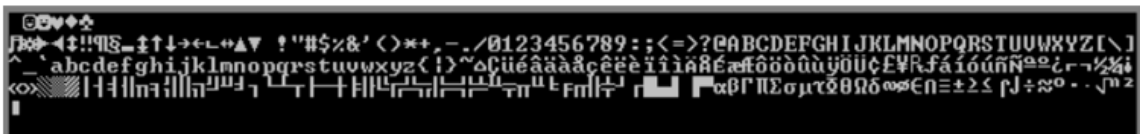
Write a program to convert following code into assembly program.

```
IF AX < BX
  THEN
    put 0 in AX
  ELSE
    IF BX < CX
      THEN
        put 0 in BX
      ELSE
        put 0 in CX
      END_IF
    END_IF
  END_IF
```

Write a program to display "o" if AL contains **1** or **3**; if AL contains **2** or **4**, display "e"..

➤ Write an assembly program to print all the ASCII code from 0 to 255.

Sample Output



Put the sum of the first 50 terms of the arithmetic sequence 1, 5, 9, 13, ... in DX. **Hints:** Employ LOOP instructions to do the following.

10.

Put the sum $100 + 95 + 90 + \dots + 5$ in AX. **Hints:** Employ LOOP instructions to do the following.

11.

Read a character and display it **50** times on the next line. **Hints:** use **LOOP** instructions and put **cx = 50**

Sample Output

Enter a character: d

dd
dddd

Thank you.

12.

Write a program to display a "?", read two capital letters, and display them on the next line In alphabetical order. Hints: use cmp, jb, xchg

Sample Output

Enter character: CB

BC

Thank you.

13.

Take an input character from user. Check it for letter and convert upper to lower or lower to upper using logical instructions.

14.

Take an input character from user. Check it for number and find whether it is odd or even using TEST instruction.

15.

- Suppose the register ax = 5, bx = 6, Swap the numbers of ax and bx so that ax gets 6 and bx gets 5. use the concept of Stack. Push and Pop instructions must use.

- Before Swap:

registers		
	H	L
AX	00	05
BX	00	06

- After Swap :

registers		
	H	L
AX	00	06
BX	00	05

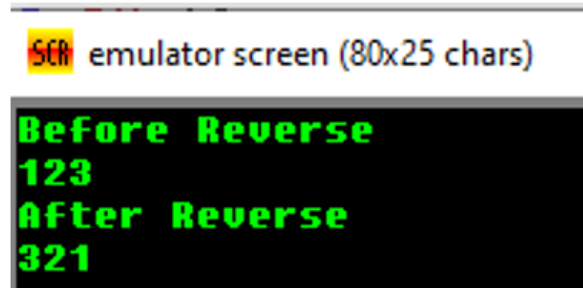
16.

- Suppose that **AX= 1234h**, **BX= 5678h**, **CX = 9ABCh**, and **SP= 1 00h**. Write an assembly program to find out the contents of **AX**, **BX**, **CX**, and **SP** after executing the following instructions:

```
PUSH AX
PUSH BX
XCHG AX,CX
POP      CX
PUSH AX
POP BX
```

17.

- Reverse three characters 123, output should look like as follows:
Hints, use the concept of push and pop.

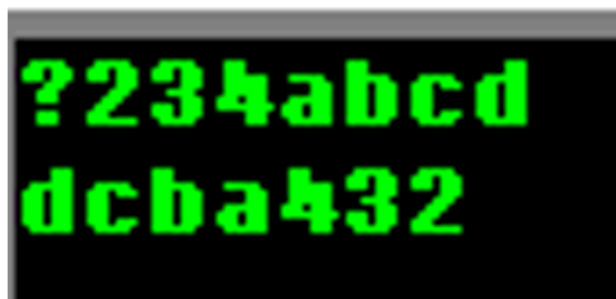


emulator screen (80x25 chars)

```
Before Reverse
123
After Reverse
321
```

18.

- Take a string from user. Once user hits enters reverse the string given by the user. Must take input from user. Hints: Use the concept of push and pop.



```
?234abcd
dcba432
```

19.

Write a program that to that has three procedures named **MAIN**, **Second** and **Third**. Print one greetings from each of the procedures.

Sample execution:

Hi, I am from **Main** Procedure

Hi, I am from **Second** Procedure

Hi, I am from **Third** Procedure

20.

Write a program that prompts the user to enter a character, and on subsequent lines prints its ASCII code in binary, and the number of 1 bit in its ASCII code. Hint: Use 2 Procedures **MAIN** and **TASK**

Sample execution:

TYPE A CHARACTER: **A**

THE ASCII CODE OF A IN BINARY IS: **01000001**

THE NUMBER OF 1 BITS IS: **2**

21.

Write a program that prompts the user to enter a character, and on subsequent lines prints its ASCII code in binary, and the number of 0 bit in its ASCII code. Hint: Use 2 Procedures **MAIN** and **TASK**

Sample execution:

TYPE A CHARACTER: A

THE ASCII CODE OF A IN BINARY IS: **01000001**

THE NUMBER OF **0** BITS IS: **6**

22.

Write a program that gives the following output in DOS. Hint: Use 2 Procedures **MAIN** and **TASK**

Welcome User!!

Please User Enter Letter 1:

J

Please User Enter Letter 2:

F

Please User Enter Letter 3:

K

Output:

J

F

K

23.

Write a program that gives the following output in DOS. Hint: Use 2 Procedures **MAIN** and **TASK**

```
Welcome To My Calculator!!  
Please User Enter A Hex Digit (A-F):  
A  
In Decimal it is: 10  
Thank You for using my Calculator  
Have a good day
```

24.

Write a program to convert small letter into capital letter. Hint: Use 2 Procedures **MAIN** and **TASK**

25.

Write a program that gives the following output in DOS. Hint: Use 2 Procedures **MAIN** and **TASK**

```
Welcome To My Calculator!!  
Please User Enter Digit 1:  
2  
Please User Enter Digit 2:  
3  
Output:  
5  
Thank You for using my Calculator
```

26.

Write a program that gives the following output in DOS. Hint: Use 2 Procedures **MAIN** and **TASK**

```
Welcome To My Calculator!!
Please User Enter Digit 1:
5
Please User Enter Digit 2:
2
Output:
3
Thank You for using my Calculator
```

27.

Write an assembly language program that takes binary inputs and gives output in reverse order. After obtaining the reversed binary output, count the number of '1's and '0's in it.

Sample Output:

```
Ente a binary number: 11010010
Reversed binary: 01001011
Number of 1's: 4
Number of 0's: 4
```

28.

Write an assembly language program to take input of a 3-character password. Each character must be a valid hexadecimal digit (0-9 or A-F). If any character is invalid, the program should display an error message and prompt the user to re-enter the password.

Sample output:

```
Enter a 3-character (hex) password: 1FZ
Invalid input!
Enter a 3-character (hex) password: A2B
Password accepted.
```

29.

Write an assembly program a) that takes the binary input (maximum 8 input) and b) counts the number of 1 bit, c) Print the count value, and d) if the binary number has odd number of 1 program prompts "ODD", otherwise program prompts "EVEN".

Sample Output:

```
USER INPUT: 10110110
NUMBER OF 1 BITS: 5
OUTPUT: ODD
```

30.

Write algorithm and assembly code for Hexadecimal input and Hexadecimal Output.